

Melissa Gibney

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EDUCATION

Carnegie Mellon University

Masters of Human Computer Interaction (MHCI)

- GPA: 4.03/4.00

University of Rochester

Bachelor of Arts in Computer Science

Bachelor of Science in Audio and Music Engineering

- GPA: 3.91/4.00 (Dean's List, Cum Laude)

Pittsburgh, PA

August 2024 – August 2025

Rochester, NY

August 2019 – May 2024

August 2019 – May 2024

WORK EXPERIENCE

Carnegie Mellon University

CyLab Extern

Pittsburgh, PA

September 2025 – Present

- Implementing picoCTF's website redesign with a homepage, profile page, and learning paths.

University of Rochester

Undergraduate Teaching Assistant

Rochester, NY

August 2021 – December 2022, August 2023 – May 2024

- Helped students understand the concepts involving audio acoustics, Fourier transforms, and MATLAB code.
- Advised students on how they should proceed during assignments involving Faust, JUCE, and C++.

Emagination STEM Camps

Program Coordinator

Lancaster, PA

June 2023 – August 2023

- Taught students (ages 8 to 17) to use C#, Python, and Unreal Engine 5 in various week-long courses.
- Organized activities and ensured that students remained safe during their stay at the camp.

General Dynamics Mission Systems

Software Engineering Intern

Dedham, MA

June 2022 – August 2022

- Updated Agent Starling (GEM One testing software) to emulate 2000 devices and run on a Tomcat server.
- Documented instructions for configuring and using Agent Starling on a virtual machine.

SELECTED PROJECTS

picoCTF Capstone Project: Researched ways to improve user engagement and retention on the picoCTF cybersecurity platform as a part of a five-person team. Implemented designs as React components with Typescript—using Storybook for documentation and Docker for containerization and shareability.

Character Creator React App: Designed a web application that allows users to customize their own character using premade pixel art assets. Each design can then be exported as an SVG.

LLM-Powered Discord Chatbot: Created a chatbot that can converse with users in Discord and respond while staying in-character using an LLM.

Flute Synthesizer: Coded a synthesized flute in Max/MSP based upon research by Professor Perry R. Cook.

Robotic Drum Machine: Worked in a team of three to design and prototype a robotic drum machine that actuates solenoids according to a sequence input by the user. The user could change the tempo and select which solenoids to actuate.

SKILLS

- Programming Languages, Libraries, and Frameworks: JavaScript, Typescript, React, Vite, HTML, CSS, Python, Flask, C, C++, C#, Java, Swift, MATLAB, R, Faust, JUCE, and Max
- Development Tools: Git, GitHub, Docker, AWS, Storybook
- Design Tools: Figma, FigJam, Miro